

Complex Analysis

Grant Talbert

March 3, 2026



MA713

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Abstract

MA713: Functions of a Complex Variable I is a graduate level course in single variate complex analysis, taught by [Professor David Rohrlich](#) in the spring 2026 semester (my junior year). The main text we followed was [[GK25](#)].

Chapter 1

Fundamental concepts

1.1 Holomorphic functions

Lecture 1: Syllabus day

Tue 20 Jan 2026 09:29

the hw is calc 2. radius of convergence is 1 for both

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}.$$

seems relevant for hw maybe?

Recall that $\mathbb{R}^2 \cong \mathbb{C}$ as $\mathbb{R}$ -vector spaces via $(1,0) \mapsto 1$ and $(0,1) \mapsto i$. When is an $\mathbb{R}$ -linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $\mathbb{C}$ -linear when viewed as a map $\mathbb{C} \rightarrow \mathbb{C}$? Recall $\dim_{\mathbb{R}} \mathbb{C} = 2$ and $\dim_{\mathbb{C}} \mathbb{C} = 1$. Well a $\mathbb{C}$ -linear map $L : \mathbb{C} \rightarrow \mathbb{C}$ from a 1-dimensional space to itself is fully determined by the image $L(1)$, since $L(z) = zL(1)$. So an $\mathbb{R}$ -linear map $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is $\mathbb{C}$ -linear precisely when $L(0,1) = (0,1)L(1) = (0, L(1))$, where $L(1,0)$ is of the form $(\alpha, 0)$ and we define $L(1) := \alpha$.

This is a shitty characterization. We can try to characterize the matrix of L as an $\mathbb{R}$ -linear map. We start with a map $L : \mathbb{C} \rightarrow \mathbb{C}$. As a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ the matrix is simply given as follows. Take $L(1) = a + bi$. Then the matrix of L is $L(1,0) = (a, b)$ and $L(0,1) = i(a + bi) = -b + ai = (-b, a)$. Thus the matrix relative to the standard basis is

$$L \sim \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Thus $\mathbb{R}$ -linear maps which induce $\mathbb{C}$ -linear maps are precisely matrices of this form. Additionally, $L(z) = z(a + bi)$ for all $z \in \mathbb{C}$. Apparently this is the final version of the Cauchy-Riemann equation.

Analysis review!!! I havent taken analysis but i bet 10 bucks i know what hes about to review. Let $U \subseteq \mathbb{R}^n$ be open. Recall $f : U \rightarrow \mathbb{R}^m$ is *differentiable* at a point $x \in U$ if and only if there exists a linear transformation $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and a map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $f(x+h) = f(x) + L(h) + \|\varphi(h)\|$ for $\|h\|$ in some ε -neighborhood of 0, and

$$\lim_{h \rightarrow 0} \varphi(h) = 0.$$

. Intuitively f is well-approximated by a linear function sufficiently close to a point x . I guess owe someone 10 bucks now. If this holds, L is called the *derivative* of f at x , also called the *Jacobian* $f_*(x) = \partial_i f^j$. It is simple to show that L is unique. Recall that matrix indices are row then column.

The main subject of this course is *holomorphy* and holomorphic functions. In $\mathbb{R}^n$, it is interesting to say whether a function is differentiable at a point. In $\mathbb{C}$, this is not interesting at all and it is more interesting to consider holomorphy on open subsets.

Definition 1.1.1. Let $U \subseteq \mathbb{C}$ open. A function $f : U \rightarrow \mathbb{C}$ is called *holomorphic* on all of U if for all $z \in U$, when viewed as a map $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is differentiable at z and the Jacobian is $\mathbb{C}$ -linear, meaning

$$\begin{pmatrix} \partial_x f^x & \partial_x f^y \\ \partial_y f^x & \partial_y f^y \end{pmatrix} \text{ is of the form } \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

In particular, $\partial_x f^x = \partial_y f^y$ and $\partial_x f^y = -\partial_y f^x$. These are the *Cauchy-Riemann* equations, usually written as $f^x = u$ and $f^y = v$.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Thus definition 1.1.1 can be rewritten as a function which is everywhere differentiable on U and satisfies the Cauchy-Riemann equations.

Example 1.1.2. Let $f(z) = z$, so $u = x$ and $v = y$. Then $\partial_i f^j = \delta_i^j$, which satisfies the Cauchy-Riemann equations.

Let $f(z) = z^2$. Then

$$f(x + iy) = (x^2 - y^2) + i2xy.$$

The derivatives are

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial v}{\partial x} = 2y, \quad \frac{\partial u}{\partial y} = -2y.$$

This satisfies the Cauchy-Riemann equations.

Let $e^z = e^x (\cos y + i \sin y)$. Then

$$\partial_i f^j = \begin{pmatrix} e^x \cos y & e^x \sin y \\ -e^x \sin y & e^x \cos y \end{pmatrix}.$$

Let $f(z) = \bar{z}$. Then $\partial_i f^j = \sigma_z$ (easy to check) which does *not* satisfy Cauchy-Riemann.

Lecture 2: The complex differential operators

Thu 22 Jan 2026 09:32

By convention, we often identify the Jacobian matrix f_* of a holomorphic function f with the single complex number $f_*(1) \cdot (1, i)$. As explored last time, we can precisely recover f_* from such a number.

We now want to work in a more natural basis of differential operators for our tangent space, in particular ∂_z and $\partial_{\bar{z}}$. We claim these are indeed linearly independent, and thus form a basis since they are of the same cardinality as ∂_x, ∂_y . We define them as follows.

Definition 1.1.3. Define the differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

This makes sense for several reasons.

Proposition 1.1.4. *A function $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$ satisfies the C-R equations if and only if $\partial_{\bar{z}}f = 0$.*

Proof. Let $f = u + iv$. Then

$$\begin{aligned}\frac{\partial f}{\partial \bar{z}} &= \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} \right) \\ &= \frac{1}{2} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) + \frac{i}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right).\end{aligned}$$

By linear independence of $\{\partial_x, \partial_y\}$, this is zero if and only if the C-R equations are satisfied. ■

Proposition 1.1.5. *Let $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$ be holomorphic. Then $f'(z) = \partial_z f|_z$.*

Proof. This follows immediately by the calculation done in the proof of proposition 1.1.4. ■

We then see by example 1.1.2 and proposition 1.1.5 that $\partial_z \bar{z} = 0$ and $\partial_{\bar{z}} z = 0$. We also have $\partial_z z = \partial_{\bar{z}} \bar{z} = 1$.

We now define a non-standard notion introduced by the book.

Definition 1.1.6. Let some $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We will call f C^1 -holomorphic if f is C^1 as a map $U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$, and $\partial_{\bar{z}}f = 0$.

The only difference between definition 1.1.1 and definition 1.1.6 is that a C^1 -holomorphic function has *continuous* partials. The more classical notion is the weaker notion. However, we will eventually see that C^1 -holomorphic functions are equivalent to holomorphic functions. We will more precisely see that holomorphic functions are C^ω .

There is an even more traditional way of saying something is holomorphic.

Definition 1.1.7. Let $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is *holomorphic* if for all $z \in U$, the limit

$$f'(z) := \lim_{h \rightarrow 0 \text{ in } \mathbb{C}} \frac{f(z+h) - f(z)}{h}$$

exists.

I went afk for a bit but we ended up with this being equivalent to

$$f'(z) = \lim_{\mathbb{R} \ni t \rightarrow 0} \frac{f(z+it) - f(z)}{it}.$$

Since we are able to formulate derivatives in terms of the same formula used in calculus, the same proofs are used to prove derivative rules for holomorphic functions, such as

$$(fg)' = f'g + fg', \quad (f \circ g)' = (f' \circ g) \cdot g'.$$

Definition 1.1.8. We say $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is *analytic* if it locally can be given by a power series, meaning for all $z \in U$ there exists an open neighborhood $z \in V \subseteq U$ such that f has a power series expansion around z which converges in V . We can write V as an ε -neighborhood $B_\varepsilon(z)$ for some $\varepsilon > 0$ sufficiently small such that $B_\varepsilon(z) \subseteq U$. We say f is C^ω and write $f \in C^\omega(U, \mathbb{C})$.

A major goal of the next few weeks is to show that f is holomorphic on U if and only if f is analytic on U . This will imply that f is C^∞ , and thus reconciling the definition given by [GK25].

Lecture 3: Analytic implies holomorphic

Tue 27 Jan 2026 09:34

Definition 1.1.9. Let $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is *analytic* or C^ω if for each $z \in U$, there exists an open set $x \in V \subseteq U$ such that f admits a power series expansion which converges in V .

Proposition 1.1.10. Suppose $\sum_n a_n w^n$ converges for some $\{a_n\}_{n \in \mathbb{N}} \subseteq \mathbb{C}$, $w \in \mathbb{C}$. If $z \in \mathbb{C}$ and $|z| < |w|$, then $\sum_n |a_n| |z|^n$ converges.

Proof. Suppose z exists, meaning $w \neq 0$. Then

$$|a_n| |z|^n = |a_n| |w|^n \left(\frac{|z|}{|w|} \right)^n.$$

It follows kinda easily from here by noting that $\{a_n w^n\}_{n \in \mathbb{N}}$ is bounded. ■

Corollary 1.1.11. If $\sum_n a_n (z - z_0)^n$ converges for some $z \in B_r(z_0)$, $r > 0$, then it converges absolutely in the entire ball.

Definition 1.1.12. Given a power series $\sum_n a_n (z - z_0)^n$, the *radius of convergence* is the number $R \in [0, \infty) \cup \{\infty\}$ such that the series converges for $z \in B_R(z_0)$, and diverges for $z \in \mathbb{C} \setminus \overline{B_R(z_0)}$.

Equivalently, $R = \sup \{|z - z_0| \mid \sum_n a_n (z - z_0)^n \text{ converges}\}$. Equivalently, recall that the lim sup of a sequence $\{c_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$, $c_n \geq 0$ is

$$\limsup_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \sup \{c_m\}_{m \geq n}.$$

We can then define $R = \left(\limsup_{n \rightarrow \infty} |a_n|^{1/n} \right)^{-1}$. It follows from these facts that if $a_n (z - z_0)^n$ converges, then so does replacing $a_n \rightarrow n a_n$ or a_n/n .

Theorem 1.1.13. Let f be analytic on $U \subseteq \mathbb{C}$. Then f is holomorphic on U .

Proof. This is a nonstandard proof the professor came up with. Let $z_0 \in U$; it suffices to show $\partial_z f$ exists at z_0 . Since f is analytic, there exists an open (wrt $\mathbb{C}$) neighborhood $V \subseteq U$ of z_0 . Write

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Let h sufficiently small such that the perturbation $f(z_0 + h)$ can be evaluated by this power series. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\sum_{n \geq 0} a_n h^n - a_0}{h} = \frac{\sum_{n \geq 1} a_n h^n}{h} = \sum_{n \geq 1} a_n h^{n-1}.$$

The limit of the right hand side should be a_1 . This makes sense since if we recall Taylor series, we should always find $a_n = f^{(n)}(z_0)/n!$. We need to show this limit exists, and we cannot assume power series are continuous on $\mathbb{C}$. Choose $|h| < r' < \text{radius of convergence}$. Since it converges, there is ε such that

$$\sum_{n \geq m+1} |a_n| (r')^{n-1} < \frac{\varepsilon}{2}.$$

Since the partial sums are continuous functions in h , we see that they approach their limit, so it like follows. ■

This proof was aids and I stopped following. There is a more canonical proof which also shows that

$$\partial_z f(z) = \sum_{n \geq 1} n a_n (z - z_0)^{n-1},$$

and that this also converges on an open disk that the original power series converges on.

Lecture 4: dsfghjk

Thu 29 Jan 2026 09:33

Proposition 1.1.14. *Let $\sum_n a_n (z - z_0)^n$ converge on some open disk $B_r(z_0)$. Then the function $f(z) = \sum_n a_n (z - z_0)^n$ is analytic on $B_r(z_0)$.*

Lecture 5: Harmonic functions

Tue 03 Feb 2026 09:31

Definition 1.1.15. A *harmonic function* is a C^2 function $u : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ such that the wave equation $\Delta u = \text{Div}_\omega du = 0$ is satisfied.

Remark 1.1.16. Since vector fields admit a canonical multiplication operation, and since $[\partial_x, \partial_y] = 0$,

$$\frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \frac{1}{4} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) = \frac{1}{4} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) = \frac{1}{4} \Delta.$$

Proposition 1.1.17. *The real and imaginary parts of a holomorphic function $f = u + iv$ are harmonic.*

Proof. We have

$$\Delta f = 4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} f = 4 \frac{\partial}{\partial z} 0 = 0.$$

By linearity, $\Delta u + i \Delta v = 0$. By $\mathbb{R}$ -linear independence of $\{1, i\}$ and since u, v are purely real, we have $\Delta u = \Delta v = 0$. ■

This statement has a partial converse. Locally, we can take an open set to always be simply connected, hence giving us a local converse.

Proposition 1.1.18. *Let $u : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be harmonic, with U open and simply connected. Then there exists a holomorphic function $f : U \rightarrow \mathbb{C}$ such that $f = u + iv$ for some v .*

The proof requires a lemma from multivariate calculus.

Proposition 1.1.19. *Let $P, Q : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be C^1 on an open set U , and satisfy $\partial_x Q = \partial_y P$. Then there exists a C^2 function $\phi : U \rightarrow \mathbb{R}$ such that $\partial_x \phi = P$ and $\partial_y \phi = Q$. Equivalently, $d\phi = (P, Q)$, meaning ϕ is a potential function for the exact vector field $(P, Q) \in \mathfrak{X}(U)$.*

Proof of proposition 1.1.18. Define

$$P := -\frac{\partial u}{\partial y}, \quad Q := \frac{\partial u}{\partial x}.$$

Then since u is harmonic, we have

$$-\frac{\partial P}{\partial y} + \frac{\partial Q}{\partial x} = -\left(-\frac{\partial^2 u}{\partial y^2}\right) + \frac{\partial^2 u}{\partial x^2} = \Delta u = 0.$$

Thus proposition [1.1.19](#) yields the potential function $v : U \rightarrow \mathbb{R}$ with $dv = (-\partial_y u, \partial_x u)$. Note that this yields

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \quad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}.$$

These are the C-R equations, so $f = u + iv$ is holomorphic on U . ■

Chapter 2

Complex line integrals

Let $\gamma : [a, b] \rightarrow \mathbb{C}$. Then γ is C^1 if

- $\gamma|_{(a, b)}$ is C^1 , and
- $\partial_t \gamma$ extends to a continuous function on $[a, b]$.

Now let f be holomorphic on $U \subseteq \mathbb{C}$ and let $\text{Im } \gamma \subseteq U$. Then we define

$$\int_{\gamma} f = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

We can view f as a section of the complexified cotangent bundle $T_{\mathbb{C}}^*U$. If $z = x + iy$, then we define $dz = dx + idy$. Then define the 1-form $\omega = f(z)dz$. Recall that complex differentials make sense in the complexified (co)tangent bundle, so this is indeed a 1-form. We then have that our definition above yields

$$\int_{\gamma} f = \int_{\gamma} \omega = \int_a^b \gamma^* \omega.$$

As usual, this is the same formula:

$$\int_a^b \gamma^* \omega = \int_a^b \gamma^*(f(z)dz).$$

Here we apply [Lee00] 11.25, which says if u is smooth and ω is a covector field, then the pullback along any F is $F^*(u\omega) = (u \circ F)F^*\omega$. This yields

$$= \int_a^b f(\gamma(t)) \gamma^* dz.$$

Note that $\gamma^* dz$ is a 1-form on $[a, b]$. To see what it is, we see how it acts on the vector field ∂_t . We have $\gamma^* dz(\partial_t) = dz(\gamma_* \partial_t)$. The (real) pushforward of $\gamma(t) = \gamma^x(t) + i\gamma^y(t)$ is

$$\gamma_* = \begin{pmatrix} \partial_t \gamma^x \\ \partial_t \gamma^y \end{pmatrix} = \frac{\partial \gamma^x}{\partial t} \frac{\partial}{\partial x} + \frac{\partial \gamma^y}{\partial t} \frac{\partial}{\partial y}.$$

Acting this on ∂_t yields simply γ_* . Then acting dz on this yields

$$(dx + idy) \left(\frac{\partial \gamma^x}{\partial t} \frac{\partial}{\partial x} + \frac{\partial \gamma^y}{\partial t} \frac{\partial}{\partial y} \right) = \frac{\partial \gamma^x}{\partial t} + i \frac{\partial \gamma^y}{\partial t} = \gamma'(t).$$

Since $\gamma^* dz$ is a 1-form on $[a, b]$, it must be of the form $g(t)dt$ for some g . By this computation, $\gamma^* dz = \gamma'(t)dt$. Note that this generalizes to piecewise-smooth curves γ .

Lecture 6: Complex line integrals

Tue 10 Feb 2026 09:32

Definition 2.0.1. Let $f : U \rightarrow \mathbb{C}$ be holomorphic. If $g : U \rightarrow \mathbb{C}$ is an *antiderivative* of f in the sense that $g' = f$, then we call g a *primitive* of f .

Terminology 2.0.2. Given $R \subseteq \mathbb{C}$ *closed*, we say f is holomorphic on R if it is holomorphic on some open neighborhood of R .

Theorem 2.0.3 (Goursat). *Let $R \subseteq \mathbb{C}$ be the closed rectangle $[a, b] \times i[c, d]$, and let f be holomorphic on R . Then*

$$\int_{\partial R} f = 0.$$

Proof. Choose a piecewise smooth parameterization γ for ∂R . Later we will show that integrals are homotopy-invariant, so it makes sense that any parameterization will suffice. Write $\alpha = \int_{\partial R} f$, and assume for contradiction that $\alpha \neq 0$. The boundary ∂R inherits the Stokes orientation. Subdivide R into four isotopic rectangles R_i (bisect each side of R with a horizontal/vertical line). These are submanifolds which inherit orientations via restricting the orientation form, and their boundaries inherit the Stokes orientation. These orientations conflict on the bisectors, but agree on the boundary. Hence they cancel on the interior, so

$$\left(\int_{\partial R_1} + \int_{\partial R_2} + \int_{\partial R_3} + \int_{\partial R_4} \right) f = \int_{\partial R} f.$$

We claim that there exists i with

$$\left| \int_{\partial R_i} f \right| \geq \frac{|\alpha|}{4}.$$

This follows immediately by

$$|\alpha| \leq \left| \sum_i \int_{\partial R_i} f \right| \leq \sum_i \left| \int_{\partial R_i} f \right| < |\alpha|.$$

Write $R^0 = R$ and R^1 for a choice of a subrectangle R_i with $\left| \int_{\partial R_i} f \right| \geq \frac{|\alpha|}{4}$. We can iterate this process to obtain a nested family of proper inclusions

$$R^0 \supsetneq R^1 \supsetneq \cdots.$$

Note that for all i ,

$$\left| \int_{\partial R^i} f \right| \geq \frac{|\alpha|}{4^n}.$$

Let P_n be the length (perimeter) of ∂R^n . Let D_n be the diameter of R^n , computed by taking the line of an affine line through two corners of R^n . Note that $\lim \{R^n\} \cong *$, and thus $P_n, D_n \rightarrow 0$. So for any choice $z_n \in R^n$, the sequence $\{z_n\}$ is Cauchy. Since f is holomorphic, we can fix z_0 and choose sufficiently small h such that $f(z_0 + h) = f(z_0) + f'(z_0)h + |h|\varphi(h)$, for

$$\lim_{h \rightarrow 0} \varphi = 0.$$

Some nice intuition that I didn't write down earlier: φ is the higher order terms of the Taylor expansion of f , hence we are really considering a germ of a Taylor expansion. Take z sufficiently close to z_0 such that we may write $h = z - z_0$ with $z \rightarrow z_0$. Then this yields

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + |z - z_0| \varphi(z - z_0).$$

Then

$$\int_{\partial R^n} f = f(z_0) \int_{\partial R^n} dz + f'(z_0) \int_{\partial R^n} (z - z_0) dz + \int_{\partial R^n} |z - z_0| \varphi(z - z_0) dz.$$

Since 1 has a primitive and is the integral around a closed curve, we can apply the fundamental theorem of line integrals to see that $\int_{\partial R^n} dz = 0$. For the same reason, the second integral vanishes. We then obtain

$$\int_{\partial R^n} f = \int_{\partial R^n} |z - z_0| \varphi(z - z_0) dz \geq \frac{|\alpha|}{4^n}.$$

However,

$$\int_{\partial R^n} |z - z_0| \varphi(z - z_0) dz \leq \int_{\partial R^n} |z - z_0| |\varphi(z - z_0)| dz \leq D_n \varepsilon P_n = \frac{D}{\alpha^n} \frac{P}{\alpha^n} \varepsilon$$

for sufficiently small ε and sufficiently large n . We choose $\varepsilon = \frac{|\alpha|}{P_n D_n}$ and some cancelling yields the contradiction

$$\frac{|\alpha|}{4^n} < \frac{\alpha}{4^n}.$$

■

Remark 2.0.4. This did not require that f have a primitive. We will show next time that holomorphic functions always have primitives.

Lecture 7: Primitive along a path

Thu 26 Feb 2026 09:36

Let f be holomorphic on U . We currently only have integrals defined over piecewise C^1 paths, but to take advantage of the notion of homotopy we want to define them for C^0 paths. So let $\gamma : I \rightarrow U$ be C^0 . We choose a partition $a = t_0 < t_1 < \dots < t_n = b$ such that $\gamma[t_{i-1}, t_i]$ falls inside a disk D_i for all i . Since f is holomorphic, it has a primitive on any open disc, so let F_i be a primitive of f on D_i . Since $D_i \cup D_{i+1}$ is connected, the primitives F_i, F_{i+1} differ by at most a constant. We choose these constants such that $F_i = F_{i+1}$; since n is finite we obtain that $F_i = F_j$ for all i, j . For $t \in [t_i, t_{i+1}]$, define $F_\gamma(t) = F_i(\gamma(t)) = ((\bigcup_i F_i) \circ \gamma)(t)$. We define

$$\int_\gamma f := F_\gamma(b) - F_\gamma(a).$$

Definition 2.0.5. A *primitive* F_γ along a path γ for a function $f \in \mathcal{O}(U)$ is a continuous function $F_\gamma : I \rightarrow \mathbb{C}$ with the property that for every $\tau \in I$, there is an open disk $D_\tau \subseteq U$ with $\tau \in D_\tau$ such that for t sufficiently close to τ , we have $F_\gamma(t) = F_\tau(\gamma(t))$ where F_τ is a primitive of f on D_τ .

Theorem 2.0.6. Let $f \in \mathcal{O}(U)$ and γ, δ homotopic paths in U . Then

$$\int_\gamma f = \int_\delta f.$$

Corollary 2.0.7 (Cauchy). Let $f \in \mathcal{O}(U)$ and γ be null-homotopic in U . Then

$$\int_\gamma f = 0.$$

Lecture 8: Homotopy invariance

Tue 03 Mar 2026 09:31

Theorem 2.0.8. *Let $f : U \rightarrow \mathbb{C}$ be holomorphic, and let $\gamma, \delta : I \rightarrow U$ be continuous paths in U with the same endpoints, which are homotopic with the endpoints fixed. Then $\int_\gamma f = \int_\delta f$.*

Proof. Just note that if f is holo then $f dz$ is closed, pull back along the homotopy, and throw Stokes at it. ■

Corollary 2.0.9 (Cauchy integration formula). *Let $f \in \mathcal{O}(U)$ and $z_0 \in U$ a point such that $\overline{B_r(z_0)} \subseteq U$. Then for any $z \in D := D_r(z_0)$,*

$$f(z) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Bibliography

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- [Lee00] John Lee. *Smooth Manifolds*. Springer, 2000.